

Question Paper

Exam Date & Time: 26-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

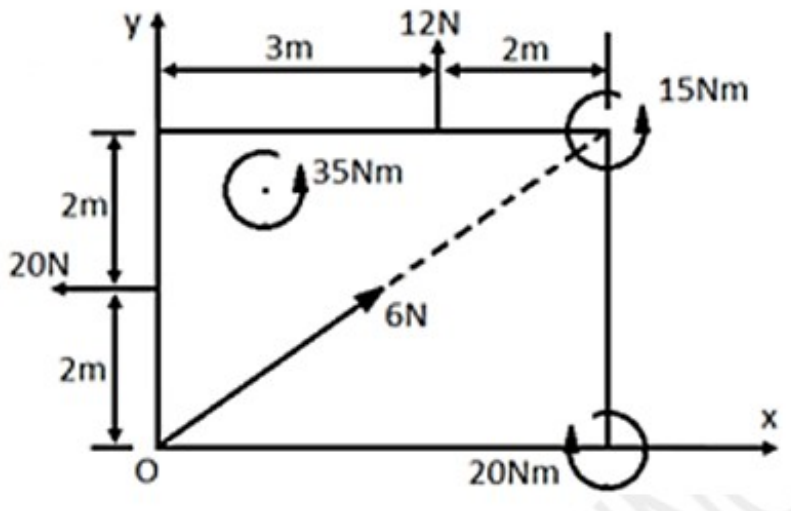
SUBJECT: CIE 1071/CIE_1071 MECHANICS OF SOLIDS

Marks: 50

Duration: 180 mins.

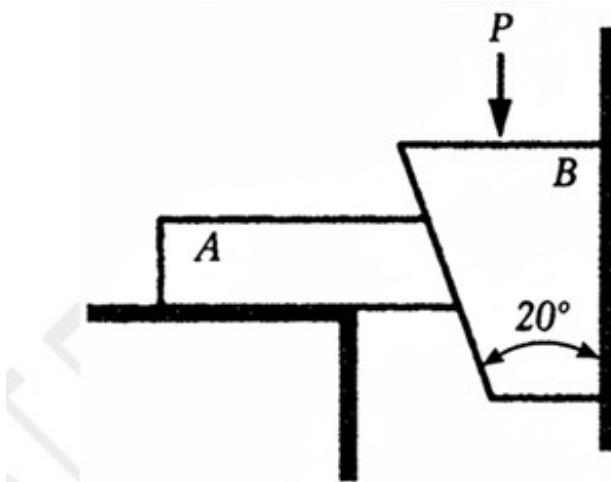
Answer all the questions.

- 1A) Determine the resultant and its position w.r.to 'O' of the non-concurrent system of forces shown in the figure. (5)

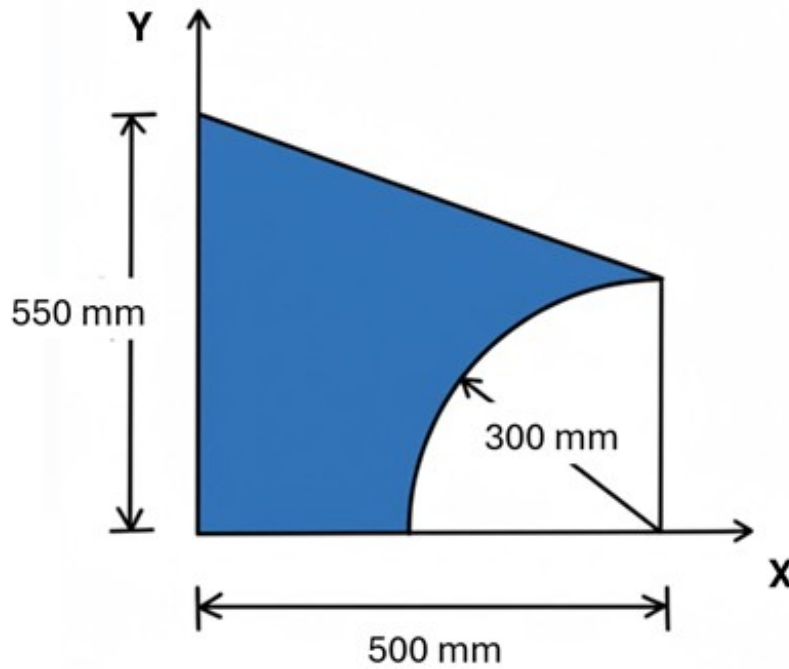


Course Outcome: "CO1" / Cognitive Level: "Application"

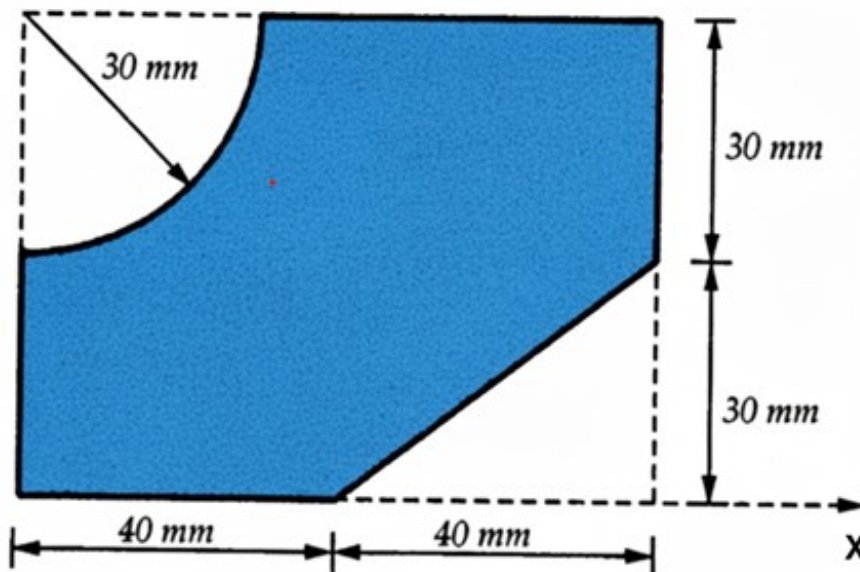
- 1B) A block A weighing 80 kN is to be moved towards left by light wedge B as shown in the figure below. Determine the necessary force P required to move block A, if angle of friction at all rubbing surfaces is 15° . (5)



- 2A) Determine the centroid for the shaded portion for the figure below w.r.t. X and Y axes. (5)

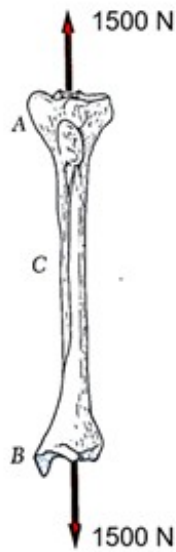


- 2B) For the shaded area shown in the figure, determine the moment of inertia and the radius of gyration about the X-axis. (5)



- 3A) A solid circular bar of length $L = 1.2$ m tapers linearly from diameter $d_1 = 30$ mm (large end) to $d_2 = 20$ mm (small end). The bar is loaded in tension with an axial load $P = 40$ kN. The material has yield strength 250 MPa and Young's modulus $E = 200$ GPa. Determine: (5)
- The axial (working) stress at the large and small ends.
 - The factor of safety against yielding based on the maximum stress.
- The total change in length (elongation) of the tapered bar under the load.

- 3B) A surgeon requires that the inner diameter at C be 10.0 mm for a medical purpose. The bone at C (5)
will carry an axial tensile load of 1500 N and the allowable average normal stress (from the strain
gauge limit) is 3.20 MPa. The modulus of elasticity of the bone material is 18 GPa, and the effective
length of the loaded bone section is 200 mm. A factor of safety (F.S.) = 1.5 must also be considered
in the design. Determine the minimum outer diameter required at C (in mm) so that average stress
does not exceed the allowable and the elongation of the bone section under the applied load.



Course Outcome: "CO4" / Cognitive Level: "Application"

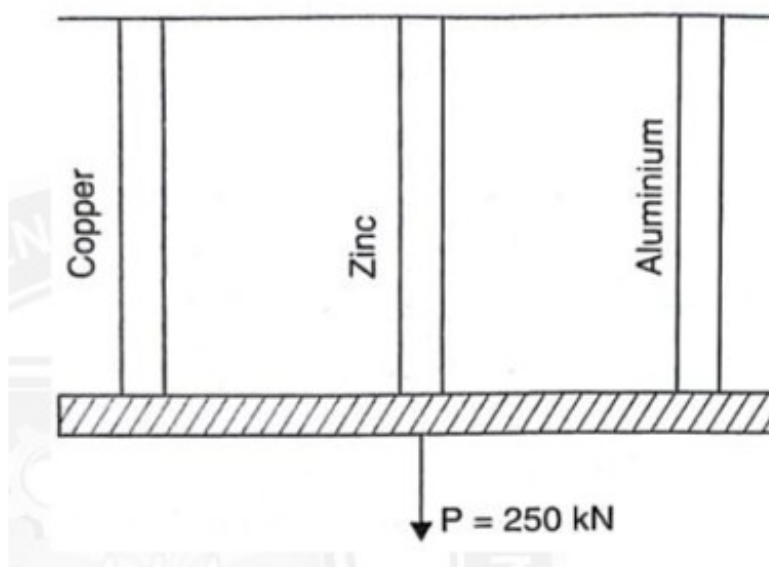
- 4A) A bar of 25 mm diameter is tested in tension. It is observed that when a load of 60kN is applied, the (5)
extension measured over a gauge length of 200 mm is 0.12 mm and the contraction in diameter is
0.0045 mm. Determine Poisson's ratio and elastic constants E, G, K.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) A cylindrical shell 3m long, which is closed at the ends, has an internal diameter of 1.5m and a wall (5)
thickness of 20mm. Determine the circumferential and longitudinal stresses induced, and the
change in the dimensions of the steel. If it is subjected to an internal pressure of 1.5 N/mm². Take
 $E = 2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio = 0.3

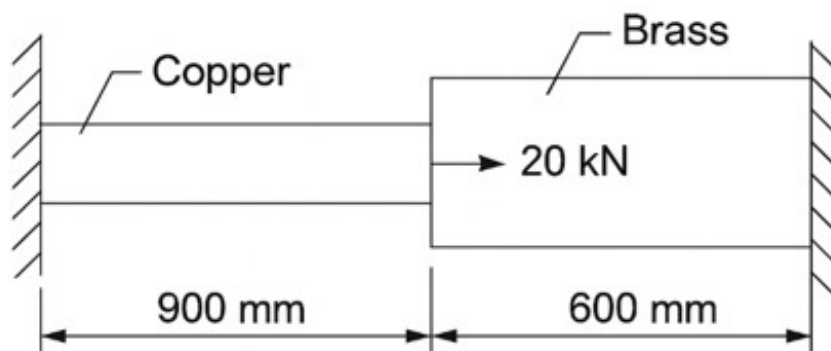
Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Three bars made of copper, zinc and aluminium are of equal length and have cross-sections 500, (5)
750 and 1000 square mm, respectively. They are rigidly connected at their ends. If the compound
member is subjected to a longitudinal pull of 250 kN. Determine the proportion of the load carried by
each rod and the induced stresses. Take the value of E for copper = $1.3 \times 10^5 \text{ N/mm}^2$, for zinc = 1.0
 $\times 10^5 \text{ N/mm}^2$ and for aluminium = $0.8 \times 10^5 \text{ N/mm}^2$.



Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) A composite bar is made by attaching a copper rod of length 900 mm and diameter 30 mm, and a brass rod of length 600 mm and diameter 45 mm, as shown in Fig. A load of 20 kN is applied at 20°C. Determine the stresses in the materials when the temperature rises to 50°C. Take $E_c = 120$ GPa, $\alpha_c = 18 \times 10^{-6}/^\circ\text{C}$, $E_b = 90$ GPa and $\alpha_a = 20 \times 10^{-6}/^\circ\text{C}$. (5)



Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CIE_1071:MECHANICS OF SOLIDS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	20
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CIE_1071:MECHANICS OF SOLIDS

#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	9	45(90%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)